

Mid - Semester Exam Rubric

Q1] Define the vectors forming the vertices:

$$\begin{aligned}\vec{BA} &= A - B = \langle 2-1, -1-0, 4-(-1) \rangle \\ &= \langle 1, -1, 5 \rangle\end{aligned}$$

$$\begin{aligned}\vec{BC} &= C - B = \langle 1-1, 2-0, 3-(-1) \rangle \\ &= \langle 0, 2, 4 \rangle\end{aligned}$$

2) In a parallelogram ABCD, the vector $\vec{AB}$ must be equal to $\vec{BC}$. Let $D = (x, y, z)$

$$\begin{aligned}\text{then } \vec{AD} &= \langle x-2, y+1, z-4 \rangle = \vec{BC} \\ &= \langle 0, 2, 4 \rangle\end{aligned}$$

$$\Rightarrow x-2 = 0 \Rightarrow x = 2$$

$$y+1 = 2 \Rightarrow y = 1$$

$$z-4 = 4 \Rightarrow z = 8$$

* Midpoint method or any other method could also be used.

(+1) mark for correctly setting up the comparison eqn $[\vec{AD} = \vec{BC}]$ (or any method)

(+1) mark for correct calculation of D

Other valid values for D include:

$$(2, -3, 0) \quad \& \quad (0, 3, 2)$$

both these solutions will be evaluated!

b) The angle at B is the angle θ b/w vectors $\vec{BA}$ and $\vec{BC}$

$$\cos(\theta) = \frac{\vec{BA} \cdot \vec{BC}}{|\vec{BA}| |\vec{BC}|}$$

$$\begin{aligned}\vec{BA} \cdot \vec{BC} &= (1)(0) + (-1)(2) + (5)(4) \\ &= 0 - 2 + 20 = 18\end{aligned}$$

$$\begin{aligned}|\vec{BA}| &= \sqrt{1^2 + (-1)^2 + 5^2} = \sqrt{1+1+25} \\ &= \sqrt{27} = 3\sqrt{3}\end{aligned}$$

$$\begin{aligned}|\vec{BC}| &= \sqrt{0^2 + 2^2 + 4^2} = \sqrt{0+4+16} \\ &= \sqrt{20} = 2\sqrt{5}\end{aligned}$$

$$\cos \theta = \frac{18}{3\sqrt{3} \cdot 2\sqrt{5}} = \frac{18}{6\sqrt{15}} = \frac{3}{\sqrt{15}}$$

(+1) mark for correct formulas

(+1) mark for correct calculation

Other valid values for $\cos \theta$:

For $D = (2, -3, 0)$

For $D = (0, 3, -2)$

$$\cos \theta = \frac{3}{\sqrt{33}}$$

$$\cos \theta = \frac{1}{\sqrt{55}}$$

Both these solⁿs will be considered only for valid 'D'.

c)
$$\text{proj}_{\vec{BC}} \vec{BA} = \left(\frac{\vec{BA} \cdot \vec{BC}}{|\vec{BC}|^2} \right) \vec{BC}$$

we know $\vec{BA} \cdot \vec{BC} = 18$ and $|\vec{BC}|^2 = 20$

$$\begin{aligned}\text{proj}_{\vec{BC}} \vec{BA} &= \left(\frac{18}{20} \right) \langle 0, 2, 4 \rangle = \frac{9}{10} \langle 0, 2, 4 \rangle \\ &= \langle 0, \frac{18}{10}, \frac{36}{10} \rangle\end{aligned}$$

(+1) mark correct formula

(+1) mark correct calculation

$$\begin{aligned}
 d) \quad \vec{BA} \times \vec{BC} &= \begin{vmatrix} i & j & k \\ 1 & -1 & 5 \\ 0 & 2 & 4 \end{vmatrix} \\
 &= i(-4-10) - j(4-0) + k(2-0) \\
 &= \langle -14, -4, 2 \rangle
 \end{aligned}$$

$$\begin{aligned}
 \text{Area} &= |\langle -14, -4, 2 \rangle| \\
 &= \sqrt{(-14)^2 + (-4)^2 + 2^2} \\
 &= \sqrt{196 + 16 + 4} \\
 &= \sqrt{216} = 6\sqrt{6} \text{ sq. units}
 \end{aligned}$$

⊕ mark correct formula

⊕ mark correct calculation

e) $\vec{BA} \times \vec{BC} (= \langle -14, -4, 2 \rangle)$ is a normal vector $\vec{n}$ to the plane.

We can use any of the points A, B, C to define the plane.

Let's use B (1, 0, -1)

The eqn of the plane is

$$\hat{n} \cdot \langle x-x_0, y-y_0, z-z_0 \rangle = 0$$

$$\Rightarrow -14(x-1) - 4(y-0) + 2(z-(-1)) = 0$$

$$\Rightarrow -14x + 14 - 4y + 2z + 2 = 0$$

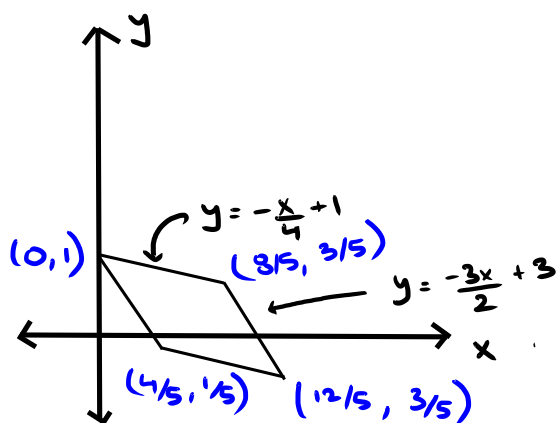
$$\Rightarrow -14x - 4y + 2z + 16 = 0$$

$$\Rightarrow 7x + 2y - z = 8$$

⊕ mark correct formula

⊕ mark correct calculation

Q2]



Given: $u = 3x + 2y$ — ①

$v = x + 4y$ — ②

$$y = -\frac{3}{2}x + 1 ; y = -\frac{3}{2}x + 3 ; y = -\frac{1}{4}x ; y = -\frac{1}{4}x + 1$$

⇓ rewriting

$$3x + 2y = 2 ; 3x + 2y = 6 ; x + 4y = 0 ; x + 4y = 4$$

⌞ ③
⌞ ④
⌞ ⑤
⌞ ⑥

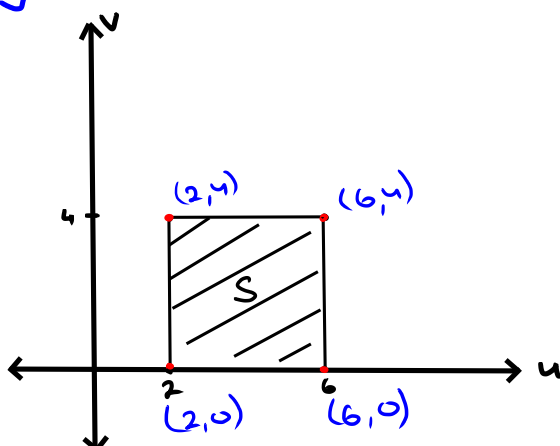
from 2, 5, 6 we get

$$v = 0 \quad \text{and} \quad v = 4$$

and from 1, 3, 4 :

$$u = 2 \quad \text{and} \quad u = 6$$

∴ Region S is a rectangle in the (uv-plane) defined by $2 \leq u \leq 6$ and $0 \leq v \leq 4$.



⊕1 mark for correct xy plane sketch

⊕1 mark for correctly rewriting the eqn

⊕1 mark for correct uv plane sketch

b)

$$u = 3x + 2y \quad v = x + 4y$$

$$\frac{\partial u}{\partial x} = 3 \quad \frac{\partial u}{\partial y} = 2$$

$$\frac{\partial v}{\partial x} = 1 \quad \frac{\partial v}{\partial y} = 4$$

$$\begin{aligned} J(x, y) &= \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} 3 & 2 \\ 1 & 4 \end{vmatrix} \\ &= (3)(4) - (2)(1) \\ &= 12 - 2 = 10 \end{aligned}$$

$$J(u, v) = \frac{\partial(x, y)}{\partial(u, v)} = \frac{1}{J(x, y)} = \frac{1}{10}$$

⊕1 mark for correct partial derivatives

⊕1 mark for correct Jacobian formula

⊕2 mark for correct final calculations

c)

The equation $(3x^2 + 14xy + 8y^2)$
can be factored as

$$(3x + 2y)(x + 4y) = uv$$

$$\therefore \iint_R (3x^2 + 14xy + 8y^2) dx dy$$

$$\Rightarrow \iint_R (3x + 2y)(x + 4y) dx dy$$

$$\Rightarrow \iint_R u.v. \, dx \, dy$$

Transforming to uv

$$\Rightarrow \iint_G uv \, J(u,v) \, du \, dv$$

$$\Rightarrow \iint_G u.v. \, \frac{1}{10} \, du \, dv$$

$$\Rightarrow \frac{1}{10} \iint_G uv \, du \, dv$$

$$\Rightarrow \frac{1}{10} \int_2^6 \int_0^4 uv \, dv \, du$$

$$\Rightarrow \frac{1}{10} \int_2^6 u \left[\frac{v^2}{2} \right]_0^4 \, du$$

$$\Rightarrow \frac{4}{5} \int_2^6 u \, du$$

$$\begin{aligned} \Rightarrow \left(\frac{4}{5} \right) \left[\frac{u^2}{2} \right]_2^6 &= \left(\frac{4}{5} \right) (18-2) \\ &= \frac{64}{5} \end{aligned}$$

(+1) mark for correctly transforming
from xy to uv plane with Jacobian

(+2) mark for correct calculations.

Q3]

$$f(x, y, z) = xy + z$$

$$g_1(x, y, z) : x^2 + y^2 - 1 = 0$$

$$g_2(x, y, z) = xz - 1 = 0$$

Lagrange Multiplier method:

The Lagrange condition is

$$\nabla f = \lambda \nabla g_1 + \mu \nabla g_2$$

$$\nabla f = (y+z, x, x)$$

$$\nabla g_1 = (2x, 2y, 0)$$

$$\nabla g_2 = (z, 0, x)$$

From which we get the system of eqⁿs

$$y+z = 2\lambda x + \mu z \quad (1)$$

$$x = \lambda(2y) \quad (2)$$

$$x = \mu(x) \quad (3)$$

$$x^2 + y^2 = 1 \quad (4)$$

$$zx = 1 \quad (5)$$

from (5) we know $x \neq 0$

from (3) we can write $\mu = 1$

from (5) we also have $z = 1/x$

substituting $\mu = 1$ in (1)

we get

$$y+z = 2\lambda x + \mu z$$

$$\Rightarrow y = 2\lambda x \quad (6)$$

Now we have

$$x = 2\lambda y \quad \text{from } (2)$$

$$y = 2\lambda x \quad \text{from } (6)$$

Substitute (6) in (2)

$$\begin{aligned} x &= 2\lambda(2\lambda x) \\ &= 4\lambda^2 x \end{aligned}$$

Since $x \neq 0$

$$\text{we get } \lambda = \pm 1/2$$

or we have $y = 0$

$\therefore$ we have three cases:

Case 1: $\lambda = 1/2$

$$\text{From } x = 2\lambda y \Rightarrow x = y$$

Substituting into (4)

$$x^2 + x^2 = 1 \Rightarrow 2x^2 = 1$$

$$x = \pm 1/\sqrt{2}$$

$\therefore$ we have two points

$$\text{for } x = 1/\sqrt{2}$$

$$y = 1/\sqrt{2}$$

and from (5)

$$z = \sqrt{2}$$

$$\text{point: } (1/\sqrt{2}, 1/\sqrt{2}, \sqrt{2})$$

$$\text{for } x = -1/\sqrt{2}$$

$$y = -1/\sqrt{2}$$

and from (5)

$$z = -\sqrt{2}$$

$$\text{point: } (-1/\sqrt{2}, -1/\sqrt{2}, -\sqrt{2})$$

Case 2 : $\lambda = -1/2$

From $x = 2\lambda y \Rightarrow x = -y$

substituting in ④

$$(-y)^2 + (y)^2 = 1$$

$$\Rightarrow 2y^2 = 1$$

$$y = \pm 1/\sqrt{2}$$

$\therefore$ similarly as above we get two points

$$pt_1 = (-1/\sqrt{2}, 1/\sqrt{2}, -\sqrt{2})$$

$$pt_2 = (1/\sqrt{2}, -1/\sqrt{2}, \sqrt{2})$$

Case 3: $y = 0$

from ④

$$x^2 + (0)^2 = 1$$

$$x = \pm 1$$

and from $2x = 1$

$$z = 1/x$$

$$\therefore pt_1 = (1, 0, 1)$$

$$pt_2 = (-1, 0, -1)$$

Now we compute the f^n evaluations at each point:

at $(1/\sqrt{2}, 1/\sqrt{2}, \sqrt{2})$

$$f(x, y, z) = 3/2$$

at $(-1/\sqrt{2}, -1/\sqrt{2}, -\sqrt{2})$

$$f(x, y, z) = 3/2$$

at $(1/\sqrt{2}, -1/\sqrt{2}, \sqrt{2})$

$$f(x, y, z) = 1/2$$

at $(-1/\sqrt{2}, 1/\sqrt{2}, -\sqrt{2})$

$$f(x, y, z) = 1/2$$

at $(1, 0, 1)$

$$f(x, y, z) = 1$$

at $(-1, 0, -1)$

$$f(x, y, z) = 1$$

$\therefore$ the maximum value is $3/2$

at $(1/\sqrt{2}, 1/\sqrt{2}, \sqrt{2})$ and $(-1/\sqrt{2}, -1/\sqrt{2}, -\sqrt{2})$

and the minimum value is $1/2$

at $(1/\sqrt{2}, -1/\sqrt{2}, \sqrt{2})$ and $(-1/\sqrt{2}, 1/\sqrt{2}, -\sqrt{2})$

saddle value at

$(1, 0, 1)$ and $(-1, 0, -1)$

Note: This problem could also be solved through the substitution method.

Basic Strategy:

- From $xz = 1 \Rightarrow z = 1/x$ ($x \neq 0$)

substitute into both the objective & cylinder constraints, reducing the number of variables.

- Then use $x^2 + y^2 = 1$ to express everything in terms of x & y & maximize/minimize $f(x, y, 1/x)$ over the curve.

Rubric :

- ① +1 mark for correctly expressing the constraints & f_n 's to be maximized / minimized
- ① +1 mark for correctly setting up the solution system
[two Lagrange eq's or the two variable constrain eq's after substitution]
- ① +1 mark for correctly writing the derived system of eq's
- ① +1 mark for correctly analysing $x \neq 0$ from $x^2=1$, $\therefore u=1$
- ② +2 mark for finding the correct candidate points
- ③ +3 marks for correct calculation and evaluation at all (and only) valid points
- ① +1 mark for stating the min & max explicitly.